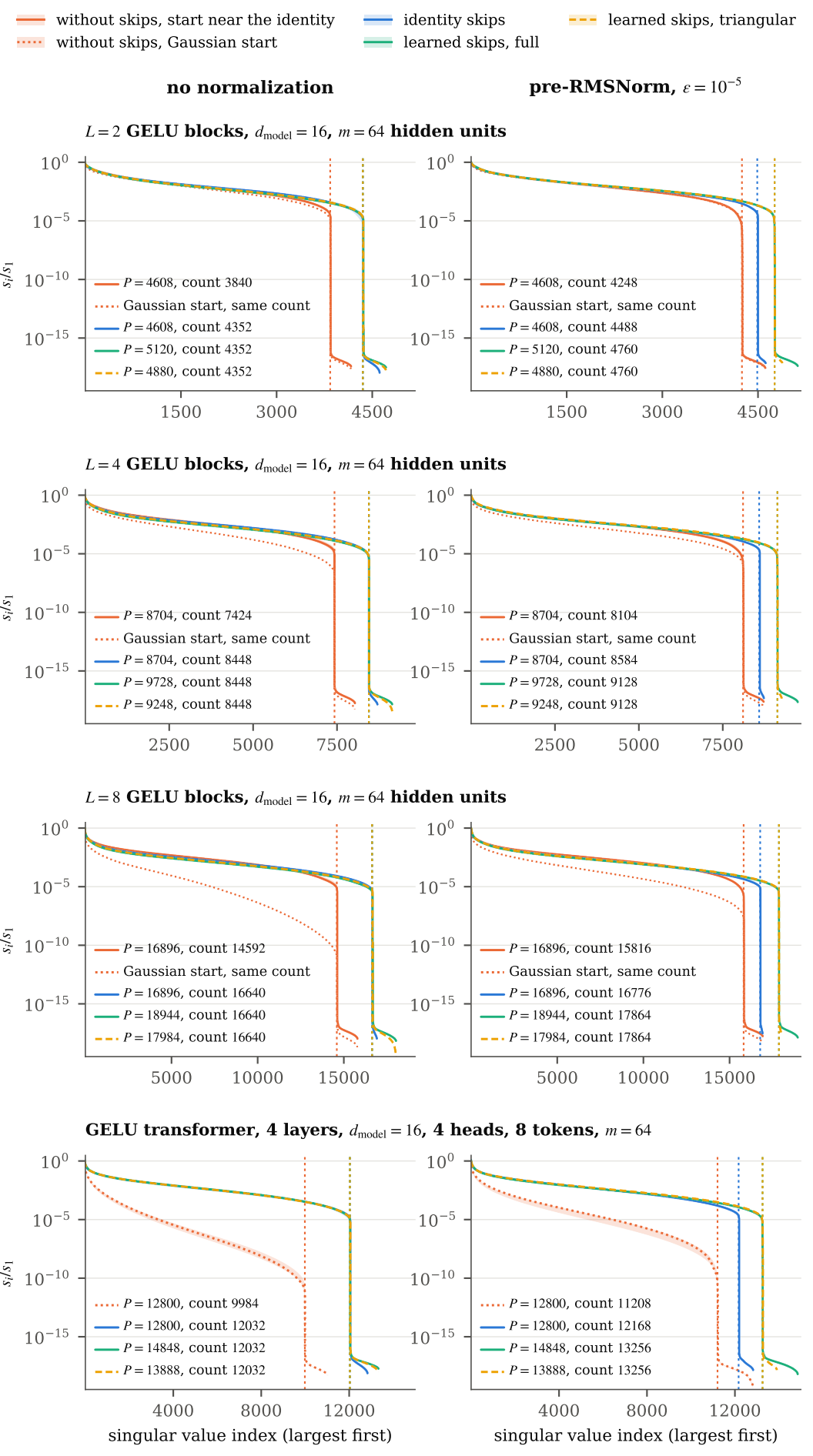

without skips, start near the identity
without skips, Gaussian start
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 2$ GELU blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
$L = 4$ GELU blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
$L = 8$ GELU blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
GELU transformer, 4 layers, $d_{\text{model}} = 16$, 4 heads, 8 tokens, $m = 64$
s_i/s_1
10^{0}
10^{-5}
10^{-10}
10^{-15}
$P = 4608$, count 3840
Gaussian start, same count
$P = 4608$, count 4352
$P = 5120$, count 4352
$P = 4880$, count 4352
$P = 4608$, count 4248
Gaussian start, same count
$P = 4608$, count 4488
$P = 5120$, count 4760
$P = 4880$, count 4760
1500
3000
4500
$P = 8704$, count 7424
Gaussian start, same count
$P = 8704$, count 8448
$P = 9728$, count 8448
$P = 9248$, count 8448
$P = 8704$, count 8104
Gaussian start, same count
$P = 8704$, count 8584
$P = 9728$, count 9128
$P = 9248$, count 9128
2500
5000
7500
$P = 16896$, count 14592
Gaussian start, same count
$P = 16896$, count 16640
$P = 18944$, count 16640
$P = 17984$, count 16640
$P = 16896$, count 15816
Gaussian start, same count
$P = 16896$, count 16776
$P = 18944$, count 17864
$P = 17984$, count 17864
5000
10000
15000
$P = 12800$, count 9984
$P = 12800$, count 12032
$P = 14848$, count 12032
$P = 13888$, count 12032
$P = 12800$, count 11208
$P = 12800$, count 12168
$P = 14848$, count 13256
$P = 13888$, count 13256
4000
8000
12000
singular value index (largest first)
singular value index (largest first)